

Predicting Sovereign Debt Crises with Macro Data, Satellite Nightlights, and News Text

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Authors

Jinen Wang, Levi Liu, Wuhao Xia, Chenyi Jiang

Abstract

Sovereign debt crises in low- and middle-income economies impose large welfare costs and are notoriously difficult to anticipate using official statistics alone, in part because such statistics are released with lags, are subject to revision, and may be strategically reported when fiscal stress is acute. This project asks whether two external data streams, VIIRS satellite nighttime lights and GDELT global news text, can improve early-warning prediction of sovereign debt crises beyond a conventional macro-only baseline. Using a country-year panel of 129 low- and middle-income economies between 2000 and 2023, we construct a forward-looking target equal to one when a sovereign debt crisis onset occurs within the next three years, and engineer a strict lag-only feature set to avoid look-ahead bias. We compare four nested XGBoost specifications, Macro Only, Macro + Satellite, Macro + Text, and All Features, on a 2019–2020 out-of-time validation period and on five rolling-origin windows, and complement them with a LASSO logistic baseline, SHAP interpretation, a regional fairness audit, an exploratory misreporting index, and a small structural signaling simulation. The results suggest that satellite nightlight features add the most stable predictive value beyond macro indicators, with Macro + Satellite achieving the highest validation average precision (0.519 vs. 0.461 for Macro Only, a 12.6% relative improvement) and also outperforming the macro baseline on the average of five rolling-origin windows (mean AP 0.366 vs. 0.345). The All Features specification posts the highest validation AUC (0.904) but lower average precision and is best treated as a robustness specification. As a supplementary methodological exercise, a leakage-safe stacking ensemble that combines XGBoost with four deep sequence learners (Appendix A) is shown not to exceed the XGBoost Macro + Satellite headline on the 2019–2020 split. The model is best interpreted as a predictive screening tool, not as a causal explanation of crises or as proof of misreporting.

1. Introduction

Sovereign debt distress remains one of the most consequential risks facing developing economies. Debt restructurings and defaults are routinely associated with sharp output losses, currency depreciation, fiscal retrenchment, and reduced public investment in health and education. Multilateral surveillance including IMF Article IV consultations and Debt Sustainability Analyses depends heavily on the macroeconomic statistics that member governments themselves report. These statistics are useful, but they are also delayed, periodically revised, and, in some cases, may be presented strategically by governments under fiscal pressure who wish to preserve market access or avoid triggering conditional lending requirements.

Traditional sovereign debt early-warning models therefore inherit the limitations of the data on which they rely. Their feature sets are dominated by official series — GDP growth, inflation, current account balance, reserves, debt-to-GDP, exchange rates — and their predictions can be systematically slow to react when those series themselves are slow or biased. A natural question for an applied machine learning project is whether external data sources, generated outside national statistical systems, can supply complementary information that is harder to suppress or revise.

This project investigates two such external sources. The first is satellite-observed nighttime light intensity from NASA’s VIIRS Black Marble products, which has been widely used as a proxy for spatially distributed economic activity (Henderson et al., 2012; Román et al., 2018). The second is the Global Database of Events, Language, and Tone (GDELT), which provides high-frequency, country-tagged event and news-text features including average tone, conflict and protest event counts, the Goldstein cooperation scale, and IMF/debt-related article mentions (Leetaru & Schrodt, 2013). Both data streams are available for low- and middle-income countries and are not produced by the governments whose statistics they complement.

We pose a single, focused research question: do satellite nightlights and news-based indicators improve sovereign debt crisis prediction beyond official macroeconomic indicators? We answer this through a nested XGBoost specification that adds satellite, text, and combined external blocks to a macro-only baseline, evaluated on a strict out-of-time validation period and a sequence of rolling-origin windows. The headline finding is that satellite features add the most stable improvement in average precision, the appropriate ranking metric for rare crisis prediction, while text features contribute useful signal that is, however, less stable across windows.

The contributions of the project are: (i) a reproducible early-warning pipeline that enforces a prediction-time information rule; (ii) a nested experimental design that isolates the marginal contribution

of satellite and text features beyond the macro baseline; (iii) a regional fairness audit and an exploratory satellite-official divergence ("misreporting") index; (iv) a clear methodological treatment of the censored 2021–2023 future period under a three-year forward-looking target; and (v) a leakage-safe stacking design for combining heterogeneous base learners without contaminating the official validation split, evaluated alongside four supplementary deep sequence models in Appendix A.

2. Research Questions

- **RQ1** Do satellite nightlights and news-based indicators improve sovereign debt crisis prediction beyond official macroeconomic indicators?
- **RQ2** Does divergence between reported GDP growth and satellite-observed activity provide useful warning signals of economic stress or possible misreporting?
- **RQ3** Does model performance vary across regions or income groups?

3. Data

3.1 Unit of Analysis and Sample

The unit of observation is the country-year. The sample is restricted, via WDI income-group codes, to low-income (LIC), lower-middle-income (LMC), and upper-middle-income (UMC) economies, with high-income economies (HIC) explicitly excluded. The temporal scope is 2000 through 2023. The merged panel covers 129 countries and 3,096 country-years. The final modeling sample is the subset of country-years for which the lag-only feature construction yields usable inputs.

3.2 World Bank WDI Macro Data

Macroeconomic indicators are downloaded from the World Bank World Development Indicators using the `wbgapi` Python client and include:

- GDP growth (NY.GDP.MKTP.KD.ZG)
- Inflation, consumer prices (FP.CPI.TOTL.ZG)
- Central government debt-to-GDP (GC.DOD.TOTL.GD.ZS)
- Current account balance, % GDP (BN.CAB.XOKA.GD.ZS)
- Total reserves in months of imports (FI.RES.TOTL.MO)
- External debt stocks, % GNI (DT.DOD.DECT.GN.ZS)
- Cash surplus/deficit, % GDP (GC.BAL.CASH.GD.ZS)
- Official exchange rate (PA.NUS.FCRF)

- GDP in constant prices and population (background controls)

Region and income-group metadata are pulled from the WDI economy table and used both for sample selection and for the regional fairness audit.

3.3 VIIRS Nighttime Lights

Country-level annual nightlight aggregates are produced via Google Earth Engine. The pipeline aggregates VIIRS imagery to country-year level and constructs a raw nightlight level (`nightlight_raw`) and annual nightlight growth (`nightlight_growth`).

Two design choices warrant emphasis. First, VIIRS-only: the project deliberately excludes the older DMSP-OLS series. DMSP and VIIRS are not directly comparable due to differences in calibration, dynamic range, and saturation; combining them without a full harmonization procedure would inject artificial breaks around the 2013–2014 sensor transition into trend-based features. Second, the project does not apply DMSP–VIIRS harmonization (e.g., Li et al., 2020), accepting a shorter satellite history as the cost of clean, internally consistent satellite features. Known measurement issues including saturation in densely lit areas, noise in small countries, and missing pixels are partially mitigated downstream by filtering for meaningful nightlight variation in the misreporting analysis.

3.4 GDELT Events and GKG News Text

Text and event features are queried from the GDELT 2.0 Event Database and Global Knowledge Graph (GKG) on Google BigQuery. At the country-year level we extract:

- **Event features:** average event tone (`avg_tone`), economic event counts (`economic_events`) and shares (`econ_event_share`), conflict event counts (`conflict_events`) and shares (`conflict_event_share`), protest event counts (`protest_events`), the Goldstein cooperation scale (`avg_goldstein`), and average event mentions (`avg_mentions`).
- **GKG article features:** article volume (`total_articles`), average document tone (`avg_doc_tone`), debt mentions (`debt_mentions`), debt article share (`debt_article_share`), IMF mentions (`imf_mentions`), IMF article share (`imf_article_share`), and tone changes (`tone_change`, `doc_tone_change`).

GDELT records are aggregated to annual frequency to match WDI and crisis-label cadence. The use of annual aggregation is deliberate: monthly interpolation against annual macro and crisis variables would create false precision and a misleading sense of within-year resolution.

3.5 Crisis Labels

Crisis labels are constructed from a curated list of sovereign debt crisis episodes drawn from Laeven and Valencia (2020), Reinhart and Rogoff (2009), and Cruces and Trebesch (2013), and IMF country reports. Each episode is encoded as a tuple (iso3c, start_year, end_year); the script generates in_crisis and crisis_onset flags at the country-year level. The labels currently end in 2023.

The prediction target is forward-looking with a three-year horizon:

$$y_{i,t} = 1 \text{ if a crisis onset occurs in country } i \text{ during years } t+1, t+2, \text{ or } t+3; 0 \text{ otherwise.}$$

This is an early-warning label: the classifier is trained to anticipate onset within a multi-year window rather than to identify contemporaneous crisis status.

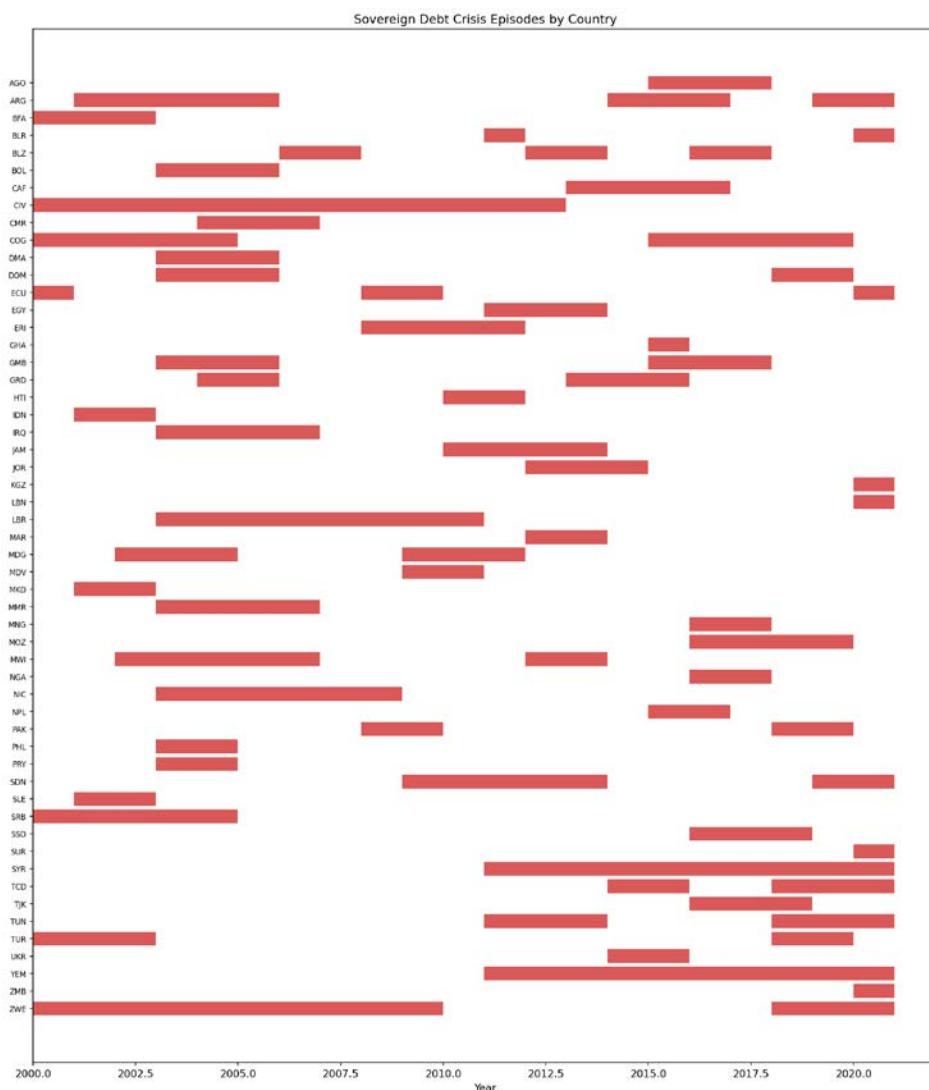


Figure 1. Crisis episodes over time (curated label set, 2000–2023).

4. Feature Engineering

4.1 Prediction-Time Information Rule

A central methodological commitment of this project is that only information available before the prediction year is allowed into the model. Feature construction enforces this through a regular-expression check that admits only columns ending in `_lagN`, `_roll_(mean|std)_Ny`, or `_change`. Current-year raw macro, satellite, or text values are never passed to any model. This eliminates a leakage channel that is easy to overlook in crisis prediction, where contemporaneous values are by construction the most predictive of contemporaneous crises.

4.2 Lagged Features

For each base variable (macro, satellite, text), the pipeline constructs:

- One-, two-, and three-year lags (`_lag1`, `_lag2`, `_lag3`).
- Three-year rolling means and standard deviations computed over prior years only (`_roll_mean_3y`, `_roll_std_3y`), implemented as `groupby("iso3c")[col].shift(1).rolling(3, min_periods=2).mean()` so that the current year is excluded from the window.
- Prior year-over-year changes (`_change`), implemented as `diff().shift(1)` to ensure the change is observed before the prediction year.
- A single interaction, `divergence_x_debt_lag1 = divergence_index_lag1 × debt_to_gdp_lag1`, motivated by the structural intuition that satellite-official divergence should matter more in highly indebted countries.

4.3 Feature Blocks

The four model specifications are built from three feature blocks.

Table 1. Feature blocks used in nested model comparison.

Block	Example variables	Intended signal	Notes
Macro (28)	<code>gdp_growth_lag1..3</code> , <code>inflation_lag1..3</code> , <code>debt_to_gdp_lag1..3</code> , <code>debt_to_gdp_change</code> , <code>current_account_gdp_lag1..3</code> , <code>reserves_months_lag1..3</code> , <code>reserves_months_change</code> , <code>external_debt_gni_lag1..3</code> , <code>exchange_rate_lag1..3</code> ,	Standard sovereign-distress fundamentals from official statistics	Baseline block

	exchange_rate_roll_(mean std)_3y, gdp_growth_roll_(mean std)_3y		
Satellite (12)	nightlight_raw_lag1..3, nightlight_growth_lag1..3, nightlight_growth_roll_(mean std)_3y, divergence_index_lag1..3, divergence_x_debt_lag1	Independently observed economic activity and divergence from reported GDP	Includes one cross-block interaction
Text (51)	Lags and rolling moments of avg_tone, avg_doc_tone, avg_goldstein, avg_mentions, total_articles, economic_events, conflict_events, protest_events, econ_event_share, conflict_event_share, debt_mentions, debt_article_share, imf_mentions, imf_article_share, tone_change, doc_tone_change	Public-information signals: media tone, conflict/protest intensity, debt and IMF salience	Sensitive to GDELT coverage variation

The All Features specification is the union of the three blocks (91–92 candidate columns, depending on training window after dropping all-missing columns).

4.4 Satellite-Official Divergence and the Misreporting Index

The merge step computes a divergence variable

$$\text{divergence_index}(i,t) = \text{nightlight_growth}(i,t) - \text{gdp_growth}(i,t),$$

and the misreporting analysis inverts this sign so that $\text{misreport_raw}(i,t) = \text{gdp_growth}(i,t) - \text{nightlight_growth}(i,t)$, with positive values indicating that reported growth exceeded what the satellite signal would suggest. Country-year observations are then converted to global z-scores after filtering to countries with moderate nightlight variability (per-country nightlight-growth standard deviation between 5 and 30) to suppress saturation and small-country noise. A country-year is flagged as anomalous when its z-score exceeds 1.5.

We emphasize that this is an exploratory diagnostic indicator. It is not a test of manipulation, it does not control for legitimate sources of divergence (sectoral composition, structural change in electrification, conflict-driven displacement), and it is best understood as a case-generation tool for further qualitative investigation rather than a validated ranking of misreporting countries.

4.5 Avoiding Look-Ahead Bias

In an early-warning setting, look-ahead bias is the dominant failure mode: a contemporaneous value of, say, `debt_to_gdp` in year t is mechanically correlated with crisis onset risk in years $t+1 \dots t+3$ simply because it shares a common time index with the target. By restricting all features to lags, prior rolling statistics, and prior changes, the project guarantees that, for a prediction made at the start of year $t+1$, every feature is observable using information realized at or before year t . Any out-of-time validation we report is therefore an honest test of forward-looking predictive performance.

5. Methods

5.1 Main Model: XGBoost

The main classifier is XGBoost. XGBoost is well-suited to this problem because (i) crisis prediction is highly nonlinear, with thresholds and interactions among debt, reserves, and growth that are difficult to capture in additive linear models; (ii) the panel is tabular and modestly sized; (iii) tree ensembles tolerate missing data and require no scaling; and (iv) the model exposes a `scale_pos_weight` parameter that is well suited to the rare-event nature of the target. Hyperparameters are fixed: 300 trees, max depth 4, learning rate 0.05, subsample 0.8, column subsample 0.8, min child weight 5. `scale_pos_weight` is set per training fold to the prevailing negative-to-positive ratio. Early stopping on the validation set is used with 30 rounds. Missing values are median-imputed within each training fold (`SimpleImputer`) and the imputer is then applied to the validation matrix.

5.2 Nested Model Specifications

Four nested specifications isolate the marginal contribution of each external block:

- **Macro Only** — the conventional baseline (28 lagged macro features).
- **Macro + Satellite** — adds the 12 lagged satellite features and the `divergence×debt` interaction.
- **Macro + Text** — adds the 51 lagged text features.
- **All Features** — the union of all three blocks.

Comparing (1) and (2) tests whether satellite features add value beyond official macro data; comparing (1) and (3) does the same for text; (4) tests whether the two external blocks complement each other.

5.3 Baseline Model: LASSO Logistic Regression

LASSO logistic regression provides an interpretable sparse baseline. Cross-validated LogisticRegressionCV with `l1_ratios=[1.0]` and a grid of regularization strengths selects an optimal penalty by 5-fold ROC-AUC. Features are standardized; class imbalance is handled by `class_weight="balanced"`. LASSO serves both as a transparent benchmark and as a feature-selection diagnostic that reveals which lagged variables survive aggressive L1 penalization.

5.4 Interpretability: SHAP

SHAP TreeExplainer is applied to the All Features XGBoost model to produce bar and beeswarm summary plots of feature attributions on the validation set. SHAP values are interpreted descriptively: they describe how much each lagged feature shifts the model's predicted log-odds (and, after the logistic link, the predicted probability) and in which direction, but they are not causal estimates. We interpret them with the same care one would apply to coefficients in a non-randomized regression.

5.5 Fairness Audit

The All Features XGBoost is trained on the standard train split; we then disaggregate validation predictions by WDI region and income group. For each region we report the number of validation observations, number of crises, AUC, and average precision. The script also computes region-specific thresholds calibrated to match the overall validation false-positive rate. These are reported as diagnostic indicators of where the global threshold under- or over-fires by region; they are not intended as deployment policy rules.

5.6 Structural Signaling Model

A two-period government–creditor signaling simulation provides theoretical motivation for the divergence–misreporting linkage. In the simulation, governments observe a true growth state θ and a debt level d ; they choose a reported value θ' to maximize a utility that trades off the cost of misreporting against the reduced borrowing rate that lenders charge in response to the report. Default occurs when servicing utility falls below a fixed default cost. The simulation is illustrative: it is not estimated from the data and does not provide causal identification. Its purpose is to check whether the empirical sign of the debt–divergence relationship is at least consistent with a strategic-reporting story.

5.7 Leakage-Safe Stacking Ensemble

Beyond the main XGBoost specification, we also evaluate a leakage-safe stacking ensemble that combines XGBoost with four deep sequence learners — LSTM (Hochreiter & Schmidhuber, 1997), GRU

(Cho et al., 2014), Temporal CNN (LeCun et al., 1998), and CNN+LSTM — through a logistic meta-model. The architecture of the four base sequence models and their individual results are deferred to Appendix A; here we describe only the ensemble design, which is the methodologically novel piece.

A standard concern when stacking many base learners is that the meta-model — typically fit by cross-validation on training-set predictions — can implicitly leak information about the validation set into the ensemble weights. We avoid this by splitting time into three disjoint windows: base learners are trained on years through 2016, the logistic meta-model is trained only on base-model predictions in the internal 2017–2018 window, and the 2019–2020 official validation set is never seen during base training or meta-training. The resulting ensemble probability for any 2019–2020 country-year is a function only of information realized at or before 2018, in exactly the same sense as the lag-only feature engineering of §4.1. Practical performance of the ensemble on this 2019–2020 split is reported in §7.8.

6. Validation Strategy

6.1 Main Out-of-Time Validation

The training period spans years through 2018; the main out-of-time validation period is 2019–2020. The validation set contains 258 country-year observations and 28 positive crisis targets, a base rate of approximately 10.9%, consistent with the rare-event nature of the problem.

6.2 Why 2021–2023 Is Right-Censored

Because the prediction target is crisis onset within the next three years, a country-year observed in calendar year t requires observed crisis labels through year $t+3$ to be evaluable. With curated labels currently ending in 2023, this means:

- 2021 requires observing onsets through 2024,
- 2022 requires observing onsets through 2025,
- 2023 requires observing onsets through 2026.

The 2021–2023 block is therefore right-censored under the current label horizon and cannot serve as a clean main test set unless labels are extended through 2026. The repository accordingly gates final-test evaluation behind an `EVALUATE_TEST_SET` flag (default `False`); any 2021–2023 results that are produced are clearly labeled as diagnostic and right-censored, not as the main reported result.

6.3 Rolling-Origin Validation

To guard against the possibility that headline performance is an artifact of one validation split, five rolling windows run and all stop strictly before the censored block:

Table 2. Rolling-origin validation windows.

Window	Train end	Val start	Val end
1	2014	2015	2016
2	2015	2016	2017
3	2016	2017	2018
4	2017	2018	2019
5	2018	2019	2020

Each specification is retrained from scratch in each window with the same hyperparameters and class-imbalance handling.

6.4 Evaluation Metrics

We report AUC, average precision (AP), and threshold-dependent precision and recall at the Youden-J optimum on the validation ROC. Average precision is treated as the headline metric because the target is rare: AUC can be high even when precision among top-ranked predictions is modest, and a useful early-warning system needs to be precise enough at high-risk thresholds to be actionable. This choice follows the methodological recommendation that precision–recall analysis is more informative than ROC analysis on highly imbalanced datasets (Saito & Rehmsmeier, 2015). We report AUC alongside AP for comparability with the broader literature.

7. Results

7.1 Main XGBoost Results

The official validation split (train $\leq$ 2018, validation 2019–2020; $N = 258$, positives = 28) yields the results in Table 3.

Table 3. Nested XGBoost validation results, 2019–2020.

Specification	Candidate features	AUC	Avg. precision	Precision	Recall

Macro Only	28	0.794	0.461	0.459	0.607
Macro + Satellite	40	0.816	0.519	0.388	0.679
Macro + Text	80	0.848	0.447	0.324	0.786
All Features	92	0.904	0.437	0.317	0.929

Three observations follow. First, the Macro + Satellite specification has the highest average precision (0.519 vs. 0.461 for Macro Only) — a 12.6% relative improvement on the rare-event metric we care most about. Second, the All Features specification has the highest AUC (0.904) and the highest recall (0.929) at the Youden threshold, but its average precision is below both Macro Only and Macro + Satellite. This pattern — high recall and AUC purchased at the cost of precision among high-risk predictions — is consistent with the model leveraging the larger text feature space to push more borderline cases above its operating threshold. Third, the Macro + Text specification improves AUC and recall over Macro Only, but its average precision is essentially flat, suggesting that the additional ranking signal from text is concentrated among medium-risk rather than top-risk observations.

We therefore designate Macro + Satellite as the preferred main model on average-precision grounds, and All Features as a robustness specification that confirms the satellite + text augmented model dominates the macro baseline on AUC.

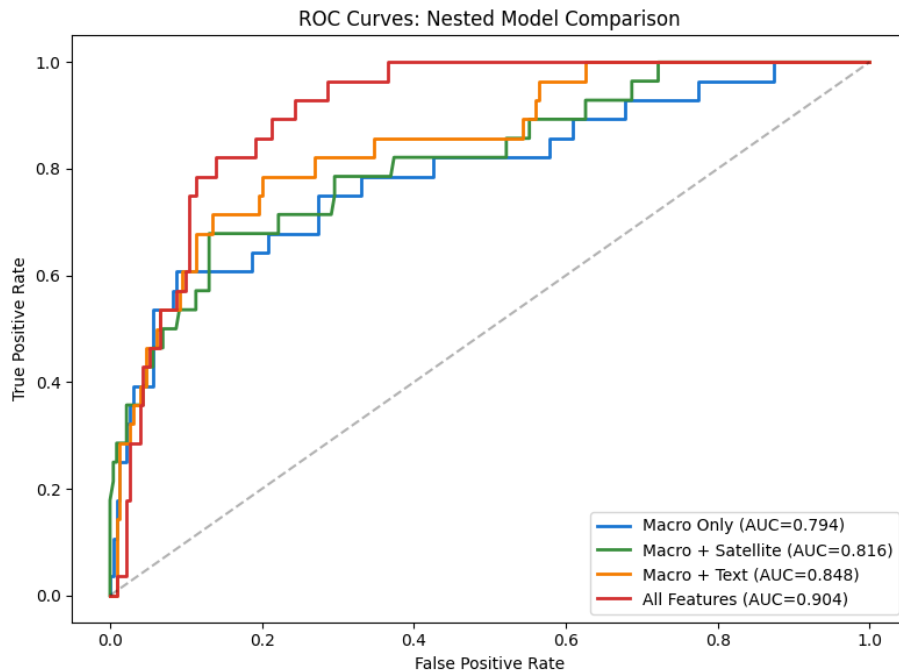


Figure 2. ROC curves for the four nested XGBoost specifications on the 2019–2020 validation set.

7.2 Rolling Validation Results

Table 4. Rolling-origin validation summary across five windows.

Specification	Windows	Mean AUC	AUC std.	Mean AP	AP std.
Macro + Satellite	5	0.759	0.049	0.366	0.182
Macro Only	5	0.759	0.045	0.345	0.137
All Features	5	0.790	0.081	0.335	0.145
Macro + Text	5	0.767	0.059	0.292	0.111

The rolling evidence is consistent with the official-split interpretation. Macro + Satellite has the highest mean average precision and an AUC indistinguishable from Macro Only, suggesting that the satellite block reliably sharpens the precision of top-ranked predictions without substantially distorting overall discrimination. The All Features specification again wins on mean AUC but falls to third on mean AP and shows the largest AUC variance across windows. Macro + Text underperforms Macro Only on both mean AP and AP stability, which is consistent with media coverage providing useful signal in some windows but degrading in others, likely reflecting heterogeneity in GDELT coverage across countries and over time.

7.3 LASSO Baseline Results

Tuned LASSO logistic regression on the same train/validation split achieves validation AUC = 0.696 and average precision = 0.202, with only 3 of 91 usable features retaining nonzero coefficients (Table 5).

Table 5. LASSO selected features (nonzero coefficients).

Feature	Block	Coefficient	Direction
reserves_months_lag1	macro	-0.097	Lower reserves → higher crisis risk
nightlight_raw_lag3	satellite	+0.046	Higher prior nightlight level associated with risk
conflict_events_lag1	text	+0.012	More prior conflict coverage → higher risk

Two interpretations follow. First, the gap between LASSO (AP = 0.202, AUC = 0.696) and XGBoost (AP up to 0.519, AUC up to 0.904) indicates that nonlinearities and feature interactions are doing real work in the boosted model, beyond what any sparse linear specification can recover. Second,

even at LASSO's aggressive penalization, the surviving features span all three blocks of macro, satellite, and text, providing a small additional piece of evidence that each block carries some signal.

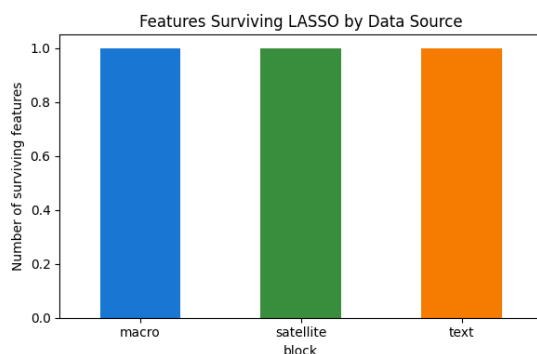


Figure 3. Surviving LASSO features by data block.

7.4 SHAP Interpretation

Inspection of the SHAP plots confirms that lagged macro variables (notably reserves months, debt-to-GDP, and external debt to GNI), satellite features (nightlight growth and divergence indices at lags), and selected text features (conflict events, IMF and debt mentions) all appear among the most influential predictors. We treat these attributions as descriptive interpretation aids and do not interpret them causally.

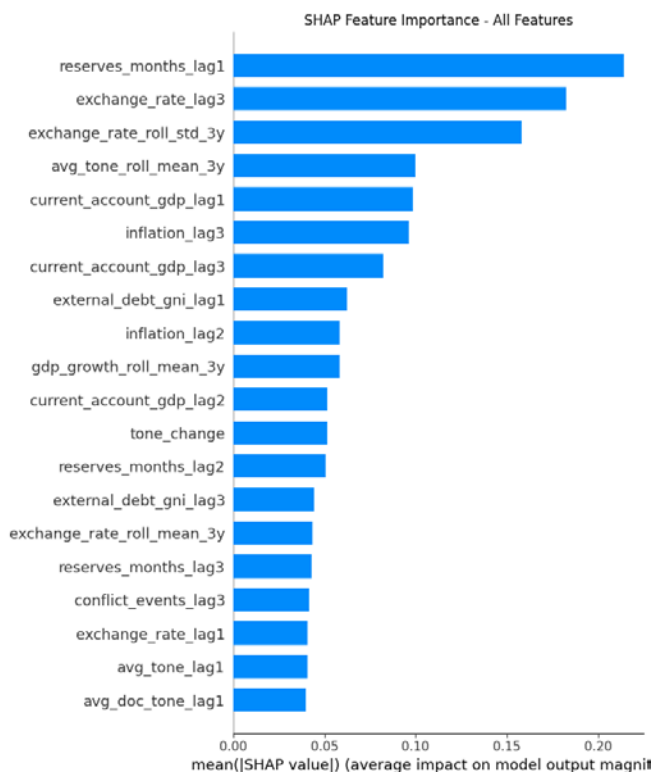


Figure 4. SHAP feature importance (mean |SHAP|) for the All Features XGBoost on the validation set.

7.5 Fairness Results

Table 6. Validation performance by WDI region (All Features XGBoost).

Region	Obs.	Crises	AUC	Avg. precision
Sub-Saharan Africa	92	5	0.947	0.431
Latin America & Caribbean	44	4	0.900	0.458
East Asia & Pacific	44	2	0.857	0.200
South Asia	12	6	0.833	0.882
Middle East & North Africa	30	5	0.888	0.486
Europe & Central Asia	36	6	0.928	0.663

The regional AUC gap is 0.114 (Sub-Saharan Africa minus South Asia), well below the 0.15 threshold the audit script flags as a warning. The lowest AP appears for East Asia & Pacific, which has the smallest number of validation crises (2). South Asia's AP of 0.882 reflects an unusually high regional crisis rate (50%) on only 12 observations and should not be interpreted as evidence that the model performs especially well in that region. By income group, validation AUC is 0.865 for LMC (n = 100) and 0.947 for UMC (n = 108).

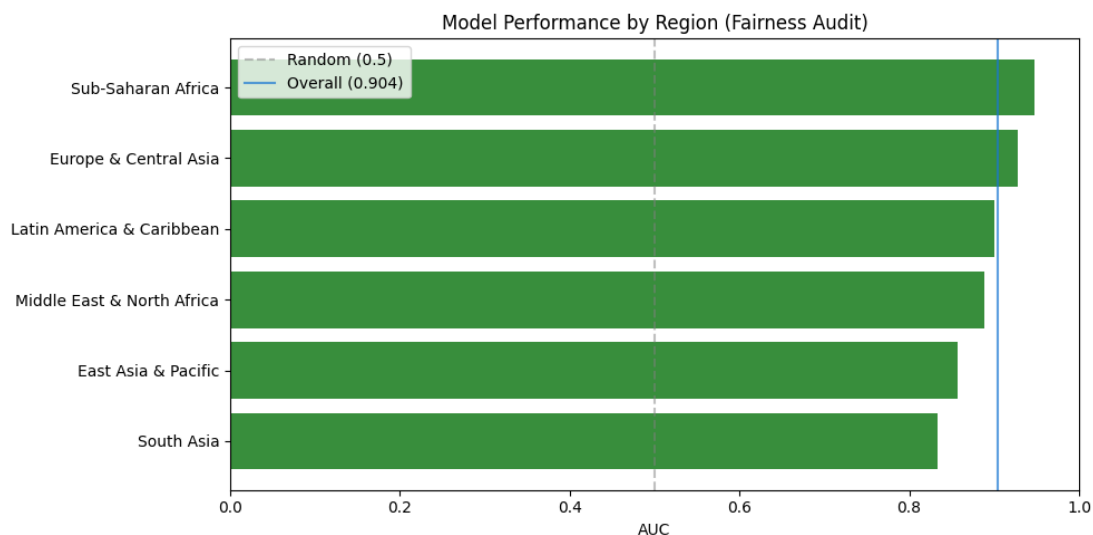


Figure 5. Regional validation AUC with overall AUC marked.

7.6 Misreporting Index Results

The exploratory misreporting index ranks 36 countries by anomaly rate (the share of years in which their satellite-vs-reported divergence exceeds the global z-score threshold of 1.5). After filtering for meaningful nightlight variability, the most-flagged cases are Moldova and Bangladesh, each with one suspicious year out of six observed years (anomaly rate ≈ 0.167). No country in the filtered sample is flagged in more than 30% of observed years. Two countries (Ecuador and Lebanon) cross the t-statistic significance threshold of $|t| > 1.96$ for a country-mean test against zero, but in both cases the divergence is negative (reported growth below nightlight growth), the opposite sign from over-reporting and likely reflecting rapid electrification or sectoral change rather than statistical concealment. Moreover, after a Bonferroni correction for 36 simultaneous country tests (corrected threshold $|t| > 2.96$), neither of these two cases would survive, reinforcing the interpretation of the index as a screening diagnostic rather than a confirmatory test.

We report this index as a case-generation tool: it identifies a small number of country-years for further qualitative investigation, but it does not establish manipulation. Legitimate reasons for divergence, sectoral composition, structural electrification trends, conflict, refugee movements, are not controlled for, and the satellite signal itself is noisy at small-country scales.

7.7 Structural Model Results

Table 7. Structural signaling model — simulated and empirical moments.

Quantity	Value
Simulated debt–divergence correlation	−0.037
Empirical debt–divergence correlation	−0.133
Simulated divergence among defaulters	0.161
Simulated divergence among non-defaulters	0.144

Both the simulated and empirical debt–divergence correlations are mildly negative, and simulated defaulters exhibit slightly larger divergence than non-defaulters in the same direction. We interpret this as directional consistency between the strategic-reporting story and the data, not as causal evidence. The magnitudes are small and the model is not estimated.

7.8 Stacking Ensemble Result

The leakage-safe stacking ensemble described in §5.7 underperforms the best single model on the 2019–2020 split. The base learners themselves, including the four deep sequence models, are reported in Appendix A. The most likely reason for the ensemble's relative weakness is the small internal 2017–2018 meta-training window, which contains too few positive examples for the logistic meta-model to learn stable ensemble weights. This is itself an informative negative result: at the sample sizes typical of country-year sovereign-debt panels, the cost of reserving a clean meta-training window can outweigh the benefit of combining base learners. We retain XGBoost Macro + Satellite as the headline specification and treat the stacking result as a methodological robustness check.

8. Discussion

RQ1 External features beyond macro indicators.

The evidence across the official validation split, rolling windows, and the LASSO baseline is consistent: satellite nightlight features add the most stable predictive value beyond official macroeconomic indicators. On the metric most appropriate for a rare-event problem (average precision), Macro + Satellite outperforms Macro Only both on the 2019–2020 split (0.519 vs. 0.461) and on the rolling-window mean (0.366 vs. 0.345). Text features carry useful signal in some windows, as they raise validation AUC from 0.794 to 0.848 and recall to 0.786 in the official split, but their average precision is not a reliable improvement and their cross-window performance is the noisiest of the four specifications.

RQ2 Divergence as a warning signal.

Satellite-official divergence is informative as a feature in the model: divergence-related variables appear in the satellite block whose addition raises validation average precision from 0.461 to 0.519 (a 12.6% relative gain), and the empirical debt–divergence correlation (-0.133) is directionally consistent with the strategic-reporting signaling simulation (-0.037). As a stand-alone country-level misreporting index, however, it is far weaker: in the 36-country filtered sample no country exhibits anomaly rates above 17% (Moldova and Bangladesh tie at 0.167), and only two cases (Ecuador and Lebanon) cross the conventional $|t| > 1.96$ threshold, neither of which would survive a Bonferroni correction for 36 simultaneous tests, and both with the opposite (negative) sign expected from over-reporting. We therefore characterize divergence as informative when used as a feature inside a multi-variable model, but not validated as a stand-alone misreporting indicator. The index is best framed as a screening and case-generation diagnostic for further qualitative investigation, not as proof of manipulation.

RQ3 Performance across regions and income groups.

Validation AUC ranges from 0.833 to 0.947 across six regions, with an AUC gap of 0.114. By income group, UMC AUC (0.947) is moderately higher than LMC AUC (0.865). These results are informative but should be treated as diagnostic rather than deployment-ready: validation subgroup samples are small (12 to 92 observations per region; only 2 to 6 crises per region), and regional thresholds calibrated on this many observations would have wide uncertainty bands.

Policy implications

The model is best read as a screening tool that complements, rather than replaces, official surveillance. Flagged country-years can prioritize where IMF, MDB, or rating-agency analysts focus more intensive review. Satellite-official divergence is most useful as a flag prompting analysts to check why reported and observed activity diverge in a particular country-year, with attention to legitimate non-statistical explanations. Because average precision in the 0.4–0.5 range and precision in the 0.3–0.5 range mean that more than half of high-confidence model alerts will be false positives, automatic decision-making like triggering surveillance escalation purely from the model is not warranted.

9. Limitations

- **Right-censoring of 2021–2023.** Because crisis labels currently end in 2023 and the target spans three years forward, the 2021–2023 block cannot serve as a clean main test set.
- **Crisis label uncertainty.** Sovereign debt crisis boundaries are inherently debatable, especially for prolonged restructurings. Using a curated episode list introduces label noise and possible omitted episodes.
- **VIIRS-only satellite history.** Restricting to VIIRS sacrifices roughly a decade of pre-2012 nightlight history, reducing training depth and constraining the rolling-window count.
- **No DMSP–VIIRS harmonization.** Implementing a published harmonization could partially recover the lost history but was deferred to future work.
- **Nightlight saturation and missingness.** Nightlights saturate in highly lit areas and are noisy for small countries; the model inherits residual measurement error.
- **GDELT coverage bias.** Media volume varies sharply by country and language coverage. The text block should be interpreted as a measure of media salience, not ground truth.
- **Annual aggregation.** Annual frequency may obscure within-year dynamics, particularly for fast-moving currency or political crises.
- **Small-sample rare-event problem.** With ~28 positives in the validation set, all metrics carry meaningful variance; this is reflected in the rolling-window AP standard deviations (0.11–0.18).

- **Subgroup sample sizes.** Regional fairness metrics are computed on subsets as small as 12 observations and 2 crises; threshold recommendations are diagnostic, not deployable.
- **Predictive, not causal.** All results including SHAP attributions, the misreporting index, and the structural simulation are predictive or descriptive. Nothing in the analysis identifies causal effects.
- **Misreporting index is exploratory.** The index does not control for legitimate non-statistical sources of satellite-official divergence and does not establish manipulation.
- **Supplementary models are not rolling-validated.** The LSTM, GRU, Temporal CNN, CNN+LSTM, and the stacking ensemble of §5.7 and Appendix A are evaluated only on the 2019–2020 official split, not across the five rolling-origin windows used for the main XGBoost specifications. Any cross-architecture comparison based on these supplementary results should therefore be treated as exploratory.

10. Future Work

- Extend curated crisis labels through 2026 to enable a proper out-of-time test on 2021–2023.
- Implement DMSP–VIIRS harmonization (e.g., Li et al., 2020) to recover a longer satellite history and a richer rolling-window evaluation.
- Validate misreporting flags against IMF Article IV reports, ROSC assessments, and country case studies.
- Calibrate operational risk thresholds for policy use (e.g., specific budgeted false-positive rates).
- Extend rolling-origin validation to the Appendix A models (LSTM, GRU, Temporal CNN, CNN+LSTM, and the stacking ensemble), and compare them against random forest and LightGBM baselines on the same five windows used for the main XGBoost evaluation.
- Incorporate higher-frequency financial-market data (sovereign CDS spreads, EMBI spreads) at observable lags.
- Improve treatment of GDELT coverage bias, e.g., by normalizing tone and event counts by country-year article volume.

11. Conclusion

This project shows that external data sources, particularly satellite nighttime lights, can sharpen the predictive content of sovereign debt crisis early-warning models beyond what official macroeconomic statistics provide on their own. On the 2019–2020 out-of-time validation period, the Macro + Satellite XGBoost specification delivers the highest average precision (0.519) and improves on the macro-only baseline both on the official split and on the average of five rolling windows. The All Features

specification achieves the highest validation AUC (0.904) and is reported as a robustness specification. Text features add useful but less stable signal. A LASSO baseline confirms that nonlinearities and interactions account for much of the gap between linear and tree-based performance, while SHAP, the regional fairness audit, the exploratory misreporting index, and a small structural signaling simulation provide interpretive context. A supplementary leakage-safe stacking ensemble combining XGBoost with four deep sequence learners (Appendix A) does not exceed the headline XGBoost Macro + Satellite specification on the 2019–2020 split; we report it as a methodological exercise rather than as a competing main model. The headline model is properly understood as a predictive screening tool, not as a causal account of crises and not as evidence of statistical manipulation by any individual government.

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Appendix A. Supplementary Experiments

A.1 Deep Sequence Models

To test whether annual country histories contain useful dynamic structure beyond the static lagged feature matrix used by the main XGBoost specifications (§5.1), we add four neural sequence learners. All four are trained on the same lag-only feature panel as XGBoost, treating each country as a sequence and each year as a time step. They are evaluated only on the official 2019–2020 split and are not subjected to the five-window rolling-origin protocol of §6.3; their results should therefore be read as exploratory.

LSTM (Hochreiter & Schmidhuber, 1997). The Long Short-Term Memory network uses PyTorch recurrent layers over short country-year sequences. Hyperparameters (sequence length, hidden units, dropout, learning rate, weight decay, batch size) are tuned by validation average precision over a compact grid; the selected configuration is reported alongside its result in Table A1.

GRU (Cho et al., 2014). The Gated Recurrent Unit is a lighter recurrent architecture than LSTM with fewer parameters per hidden unit. It is included on the prior that simpler recurrent models should generalize better than larger ones on a small annual panel.

Temporal CNN (LeCun et al., 1998). A one-dimensional convolution applied along the annual sequence axis. The intuition is that local temporal motifs including consecutive years of reserve deterioration, debt acceleration, or worsening news tone, may carry information that lag-only features only partially encode.

CNN+LSTM. A hybrid in which 1D convolution first extracts local temporal motifs and an LSTM then models their longer-range sequential composition. This is the most parameter-rich of the four and is included to test whether the added expressiveness pays off on so small a panel.

A.2 Stacking Ensemble Design

The leakage-safe stacking ensemble combining XGBoost with the four sequence models above is described in §5.7 and is not repeated here; the result is reported in §7.8 and again in Table A2 alongside the strongest single sequence model in each feature specification.

A.3 Results

Table A1 reports the tuned LSTM result within each of the four nested feature specifications used by the main XGBoost comparison (§5.2). The strongest LSTM result is Macro + Satellite (AP = 0.508), close to but below the headline XGBoost Macro + Satellite AP of 0.519. LSTM Macro + Text is also competitive (AP = 0.492), suggesting that explicit sequence modeling can extract some additional ranking signal from time-ordered text features that the static lagged representation does not capture.

Table A1. Tuned LSTM validation results, 2019–2020.

Specification	Candidate features	AUC	Avg. precision	Precision	Recall
Macro Only	28	0.785	0.425	0.467	0.500
Macro + Satellite	40	0.792	0.508	0.303	0.714
Macro + Text	80	0.845	0.492	0.276	0.750
All Features	92	0.770	0.456	0.279	0.607

Table A2 reports, for each feature specification, the strongest single model among the additional architectures (LSTM, GRU, Temporal CNN, CNN+LSTM). The strongest single result is Temporal CNN on Macro Only (AP = 0.525, AUC = 0.852), nominally above the main XGBoost Macro + Satellite AP of 0.519. We caution against reading this number as a victory: it is a single-split result on the smallest feature block, with $N = 258$ country-years and 28 positives, and it is the best of twenty independent (4 specification $\times$ 5 model) evaluations, so its multiple-comparison risk is non-trivial. Without rolling-origin replication (§6.3) we cannot tell whether this is a stable improvement or a split-specific artifact. GRU is the most consistent extension across the richer feature blocks. CNN+LSTM never exceeds GRU, which is consistent with the prior that simpler recurrent architectures are preferable when sequence length and sample size are small.

Table A2. Best single supplementary model within each feature specification.

Specification	Best model	AUC	Avg. precision	Precision	Recall
Macro Only	Temporal CNN	0.852	0.525	0.379	0.786

Macro + Satellite	GRU	0.764	0.504	0.372	0.571
Macro + Text	GRU	0.853	0.482	0.281	0.893
All Features	GRU	0.735	0.463	0.471	0.571

A.4 Interpretation

These supplementary results do not overturn the main model choice. The leakage-safe stacking ensemble (§7.8) underperforms the best single base model on the 2019–2020 split, and no individual sequence model is shown to be robustly better than XGBoost Macro + Satellite once rolling-origin variance is taken into account, and that variance is by construction not measured here. We therefore retain XGBoost Macro + Satellite as the headline specification of the project and treat Appendix A as evidence that (i) the data do contain some additional sequence structure that simple lag features only partially capture, and (ii) rigorous comparison of XGBoost against deep sequence alternatives on this kind of small, rare-event panel requires the same rolling-origin discipline applied to XGBoost in §6.3, which we leave to future work (§10).